
Preface

A statistical model is a probability distribution constructed to enable inferences to be drawn or decisions made from data. This idea is the basis of most tools in the statistical workshop, in which it plays a central role by providing economical and insightful summaries of the information available.

This book is intended as an integrated modern account of statistical models covering the core topics for studies up to a masters degree in statistics. It can be used for a variety of courses at this level and for reference. After outlining basic notions, it contains a treatment of likelihood that includes non-regular cases and model selection, followed by sections on topics such as Markov processes, Markov random fields, point processes, censored and missing data, and estimating functions, as well as more standard material. Simulation is introduced early to give a feel for randomness, and later used for inference. There are major chapters on linear and nonlinear regression and on Bayesian ideas, the latter sketching modern computational techniques. Each chapter has a wide range of examples intended to show the interplay of subject-matter, mathematical, and computational considerations that makes statistical work so varied, so challenging, and so fascinating.

The target audience is senior undergraduate and graduate students, but the book should also be useful for others wanting an overview of modern statistics. The reader is assumed to have a good grasp of calculus and linear algebra, and to have followed a course in probability including joint and conditional densities, moment-generating functions, elementary notions of convergence and the central limit theorem, for example using Grimmett and Welsh (1986) or Stirzaker (1994). Measure is not required. Some sections involve a basic knowledge of stochastic processes, but they are intended to be as self-contained as possible. To have included full proofs of every statement would have made the book even longer and very tedious. Instead I have tried to give arguments for simple cases, and to indicate how results generalize. Readers in search of mathematical rigour should see Knight (2000), Schervish (1995), Shao (1999), or van der Vaart (1998), amongst the many excellent books on mathematical statistics.

Solution of problems is an integral part of learning a mathematical subject. Most sections of the book finish with exercises that test or deepen knowledge of that section, and each chapter ends with problems which are generally broader or more demanding.

Real understanding of statistical methods comes from contact with data. Appendix A outlines practicals intended to give the reader this experience. The practicals themselves can be downloaded from

<http://statwww.epfl.ch/people/~davison/SM>

together with a library of functions and data to go with the book, and errata. The practicals are written in two dialects of the S language, for the freely available package R and for the commercial package S-plus, but it should not be hard for teachers to translate them for use with other packages.

Biographical sketches of some of the people mentioned in the text are given as sidenotes; the sources for many of these are Heyde and Seneta (2001) and

<http://www-groups.dcs.st-and.ac.uk/~history/>

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My long-suffering family deserve the most thanks. I dedicate this book to them, and particularly to Claire, without whose love and support the project would never have been finished.

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